Krona plot visualization with Qiime2

Stefan Keller

12/7/25

Video: https://www.youtube.com/watch?v=cCDJvdMcD34

Purpose: This tutorial will outline how to prepare a dataset for Qiime2 import and how to use Qiime2 for krona plot visualization.

Background: Commensal diversity within an organism plays and essential role in general biologic functionality and health. Consequently, shifts in these populations can have direct and indirect impacts on local and systemic health. To establish whether microbial shifts are associated with disease pathogenesis, a microbial population baseline abundance is needed. Qiime2 is a command line-based interface used to quantify and visualize microbial population abundance based on microbial DNA or RNA sequence analysis after library sequencing (1). Compared to stacked bar charts or heatmaps, krona plots are more powerful for intra-sample microbial diversity due to its interactive capabilities at the different biological classification levels (genus, species, etc.). This allows for visualization of how microbial populations shift if significant differences are observed for alpha and beta diversity. Alpha diversity refers to the differences in microbial populations within a sample, while beta diversity refers to differences between two or more samples (differences between two distinct sample or in one sample at two different time points) (2). Though alpha and beta diversity analysis is outside the scope of this tutorial, it is a downstream application that can be evaluated in Qiime2 following krona plot visualization.

Installation requirements:
Note: This tutorial is designed for a Unix based environment. The computer model used for this project was macOS (version 12.7.6, processor: 2.7 GHz Dual-Core Intel Core i5). The sites/codes for installing the necessary components are listed below:

a. Miniconda : https://www.anaconda.com/docs/getting-started/miniconda/install#macos-2
b. BioConda (code for installation): conda install bioconda::macs
c. Paralell (code for installation): conda install conda-forge::parallel
d. SRA-tools: https://github.com/ncbi/sra-tools/wiki/02.-Installing-SRA-Toolkit
e. Qiime2: https://docs.qiime2.org/2024.10/install/native/
    i. For krona plots, only need the "Qiime2 Amplicon Distribution" option
f. Krona plot: GitHub: https://github.com/kaanb93/q2-krona
g. Workflow/code for importing SRA dataset into Qiime2: https://github.com/stephanbitterwolf/Qiime2/blob/main/05_NCBI_to_QIIME2

Other helpful links: Qiime2 tutorial

a. This is the basic tutorial for using and becoming familiar with Qiime2 commands and data interpretation.

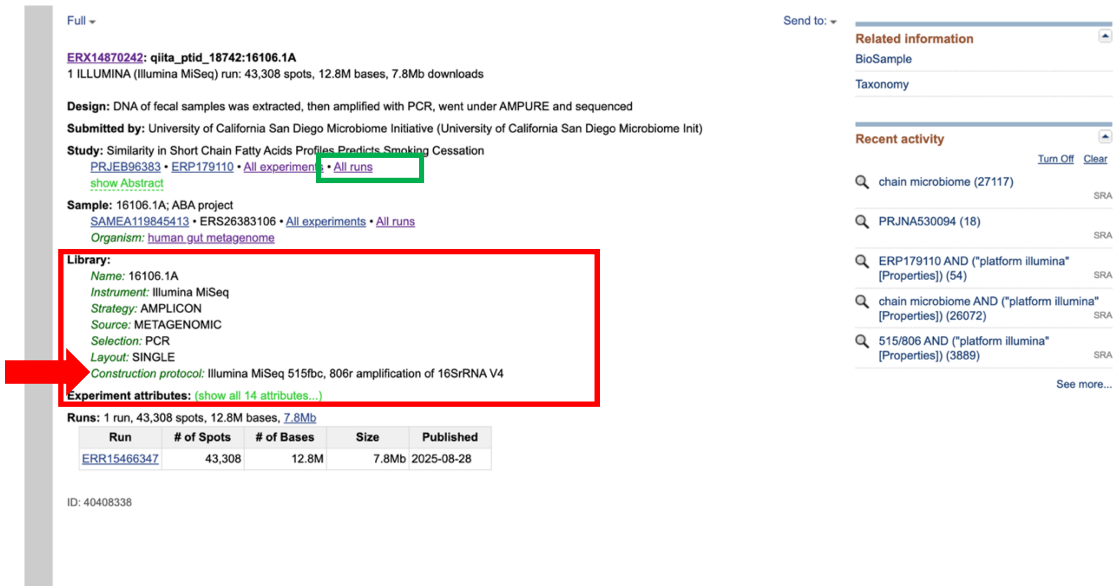
Dataset used in this tutorial (Accession: PRJEB96383):
This dataset was part of a study related to the human microbiome based on fecal microbiota samples (noninvasive measurement of microbial populations). The primary reason this dataset was selected was due to its inclusion of barcode sequences associated with each sample. As will be discussed, barcode sequences are a necessary part of the table for Qiime2 import.

Tutorial steps:

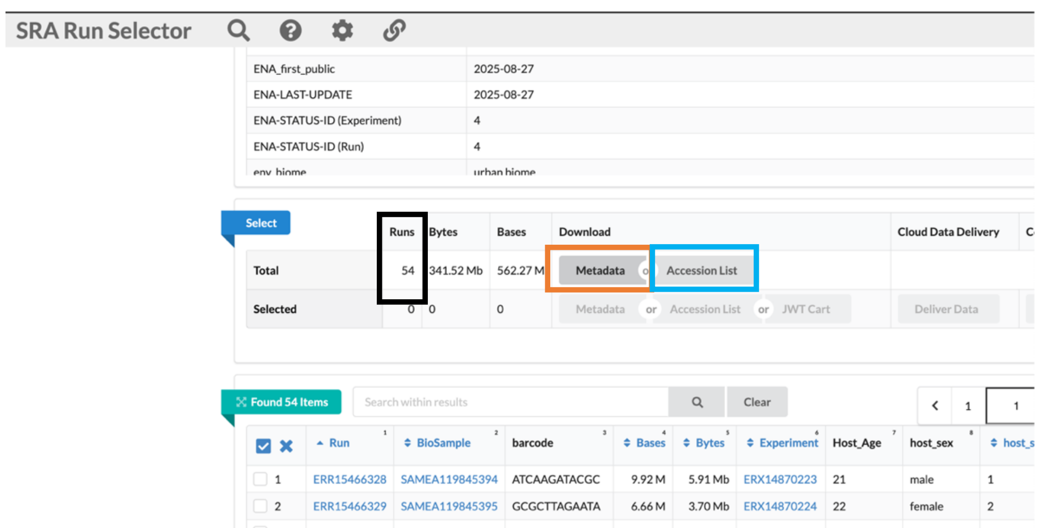
<u>Setting directories and downloading dataset</u>

1. Open terminal
2. Set directories
    a. mkdir data
    b. cd data
    c. mkdir microbiome
    d. cd microbiome
3. Downloading data from sequence read archive (SRA) files
    a. Go to https://www.ncbi.nlm.nih.gov/sra
    b. Go to search bar (set on "SRA" for search)
    c. Search for accession ID "PRJEB96383"
    d. Click on first option

    e. Click on the dataset, then under the "Study" section, click on "All runs" (green box)
        i. Under "Library": says information of the library, including whether its single or paired end reads (under "Layout"). In this dataset, also has "Construction protocol" (red arrow) which says which primer pairs and microbial variable region is being amplified for analysis



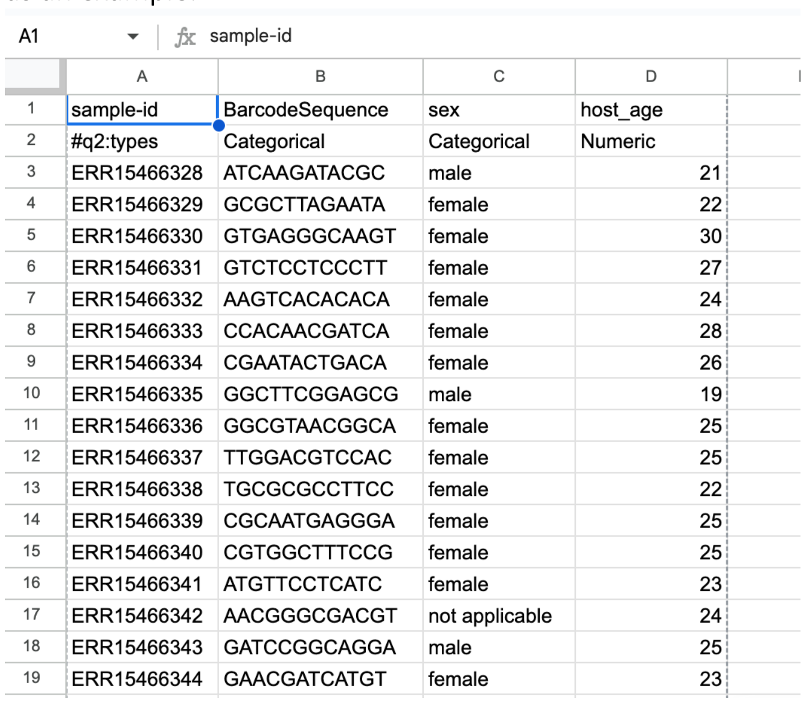
f. After clicking on "All Runs", should have a total of 54 runs (black box). Click "Metadata" to download metadata as CSV (orange box). Click "Accession list" to download run id's as a "txt" file (blue box). Save txt file to current directory as txt file. Rename as "smoke_sample.txt".



4. Reformat metadata to Qiime2 format
    a. Import metadata CSV into google sheets
    b. Reformat metadata to Qiime2 format (example of formatted spreadsheet below)
        i. Insert row below "Run", then change "Run" to "sample-id".
        ii. Under sample-id add "#q2:types"
            1. Qiime2 only recognizes columns classified as "Categorical" or "Numeric"

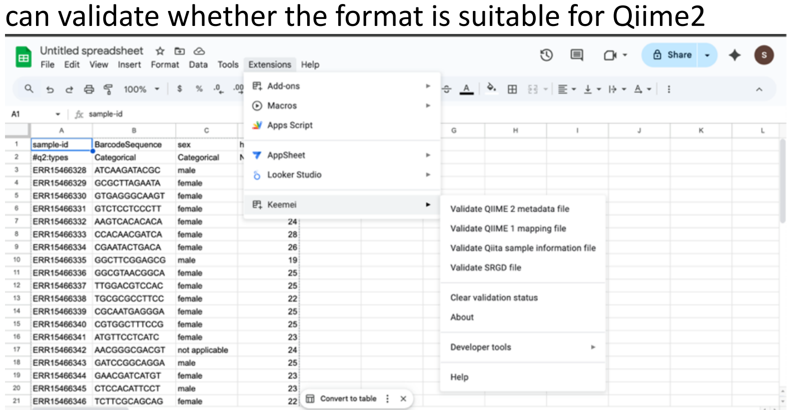
iii. Find the barcode sequence and move it to the second column. Change name to “BarcodeSequence”

1. In the #q2:types row, label as Categorical
2. This is technically all you need to make a krona plot, though other variables can be included in the metadata being imported into Qiime2 if downstream analysis (like alpha and/or beta diversity) is going to be analyzed. The current metadata has sex and host age as an example.



	A	B	C	D
1	sample-id	BarcodeSequence	sex	host_age
2	#q2:types	Categorical	Categorical	Numeric
3	ERR15466328	ATCAAGATACGC	male	21
4	ERR15466329	GCGCTTAGAATA	female	22
5	ERR15466330	GTGAGGGCAAGT	female	30
6	ERR15466331	GTCTCCTCCCTT	female	27
7	ERR15466332	AAGTCACACACA	female	24
8	ERR15466333	CCACAACGATCA	female	28
9	ERR15466334	CGAATACTGACA	female	26
10	ERR15466335	GGCTTCGGAGCG	male	19
11	ERR15466336	GGCGTAACGGCA	female	25
12	ERR15466337	TTGGACGTCCAC	female	25
13	ERR15466338	TGCGCGCCTTCC	female	22
14	ERR15466339	CGCAATGAGGGA	female	25
15	ERR15466340	CGTGGCTTTCCG	female	25
16	ERR15466341	ATGTTCCTCATC	female	23
17	ERR15466342	AACGGGCGACGT	not applicable	24
18	ERR15466343	GATCCGGCAGGA	male	25
19	ERR15466344	GAACGATCATGT	female	23

3. Optional: Can install Keemi as extension in Googlesheets, which can validate whether the format is suitable for Qiime2



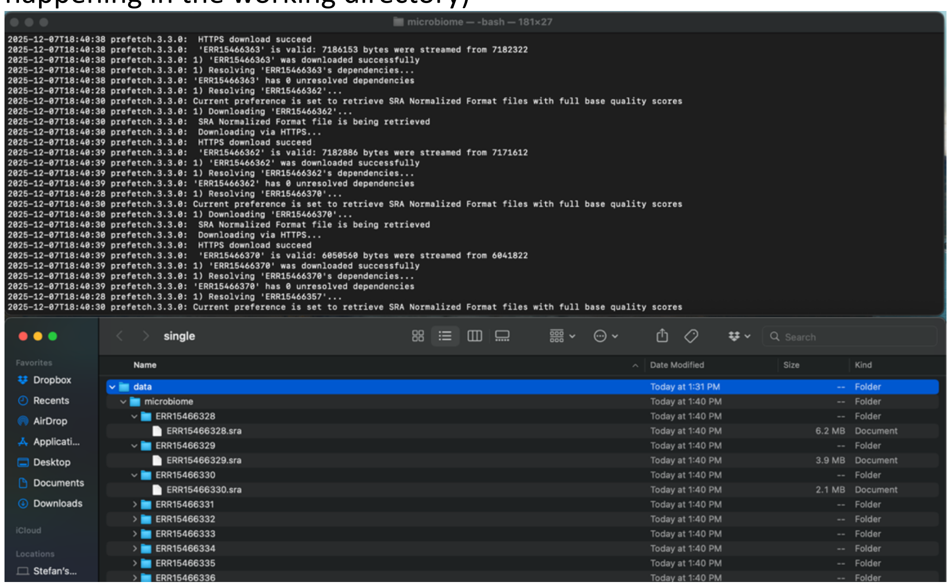
iv. Delete any columns that are not being used. Qiime2 will not be able to interpret data that is not categorized as “Categorical” or “Numeric”, and there can sometimes be 100+ columns that would need to be manually categorized before Qiime2 import. For this tutorial, since we are only using the columns “BarcodeSequence”, “sex” and “host_age”, removing extra columns is the easiest option for Qiime2 import.

v. Export as “TSV” file to current directory. Rename TSV file as “metadata.tsv”

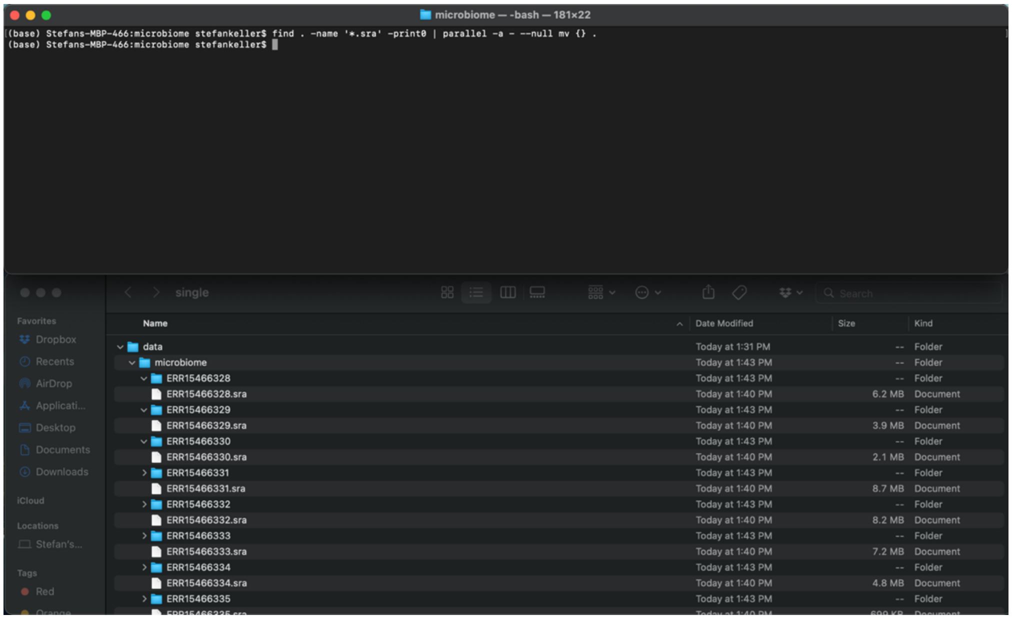
5. SRA to fastq files for Qiime2 (code modified from [here]): SRA files are compressed binary files used for storing large datasets, and is a useful format for storage in public repositories, such as NCBI. While SRA and fastq files contain the same information, bioinformatic tools requires the text-based/ human readable fastq format for subsequent downstream analysis.

   a. Download data in parallel: cat smoke_sample.txt | parallel -j0 prefetch {}

      i. Will get a bunch of folders with single SRA file (top is the terminal with resolved commands, bottom is the working directory and what is happening in the working directory)

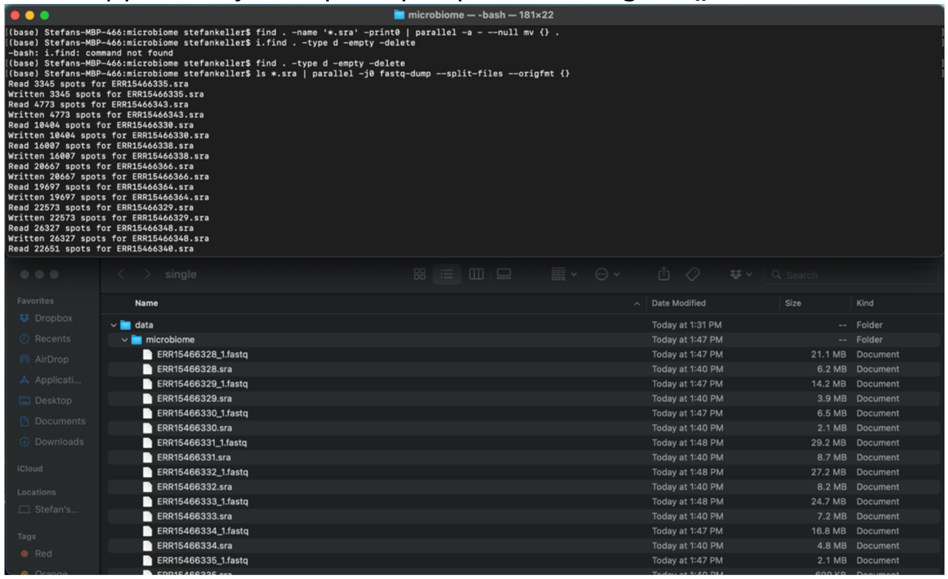


b. Move all SRA files out of their folders in the working directory in parallel:
    i. find . -name '*.sra' -print0 | parallel -a - --null mv {} .



c. Delete empty folders
    i. find . -type d -empty -delete

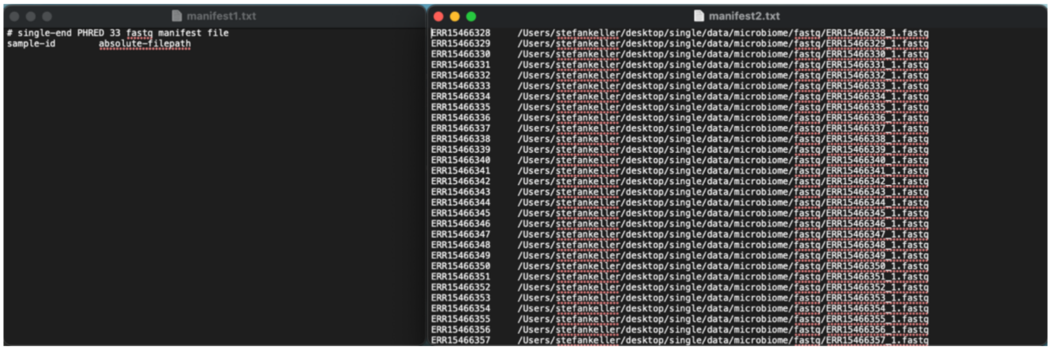
d. Convert SRA files to fastq files:
    i. ls *.sra | parallel -j0 fastq-dump --split-files --origfmt {}



    ii. Now we have both SRA files and fastq files for each sample

e. Separate SRA and fastq files into separate folders: SRA files are not needed from this point forward, but if a fastq file becomes corrupted or lost, it may be useful to have the original file available
    i. mkdir fastq | mv *.fastq fastq
    ii. mkdir sra | mv *.sra sra

6. Create manifest for Qiime2: The manifest file is important for mapping demultiplexed sequences to a single Qiime2 artifact based on the sample ID and absolute file path (1), which can then be viewed and interpreted
    a. Change to fastq directory
    b. Create txt file header
        i. echo "# single-end PHRED 33 fastq manifest file" > manifest1.txt
        ii. echo -e "sample-id\t absolute-filepath" >> manifest1.txt
    c. Create txt file with ID path separated by tabs
        i. ls *.fastq | cut -d "_" -f 1 | sort | uniq | parallel -j0 --keep-order 'echo -e "{/}\t"$PWD"/{/}_1.fastq"' | tr -d '"' > manifest2.txt
    d. When viewing manifest 1 and manifest 2, manifest 1 is essentially the column titles, and manifest 2 is the information for each column separated by tab



manifest1.txt

```
# single-end PHRED 33 fastq manifest file
sample-id	absolute-filepath
```

manifest2.txt

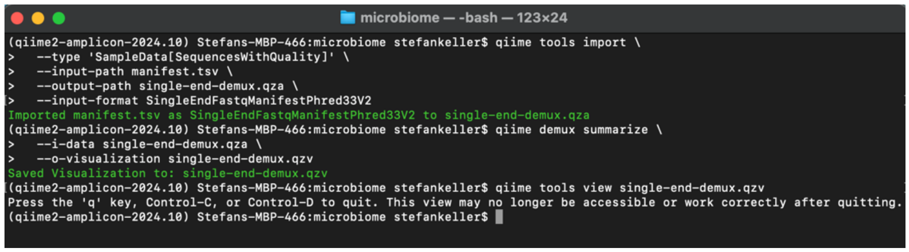
```
ERR15466328	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466328_1.fastq
ERR15466329	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466329_1.fastq
ERR15466330	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466330_1.fastq
ERR15466331	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466331_1.fastq
ERR15466332	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466332_1.fastq
ERR15466333	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466333_1.fastq
ERR15466334	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466334_1.fastq
ERR15466335	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466335_1.fastq
ERR15466336	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466336_1.fastq
ERR15466337	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466337_1.fastq
ERR15466338	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466338_1.fastq
ERR15466339	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466339_1.fastq
ERR15466340	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466340_1.fastq
ERR15466341	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466341_1.fastq
ERR15466342	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466342_1.fastq
ERR15466343	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466343_1.fastq
ERR15466344	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466344_1.fastq
ERR15466345	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466345_1.fastq
ERR15466346	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466346_1.fastq
ERR15466347	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466347_1.fastq
ERR15466348	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466348_1.fastq
ERR15466349	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466349_1.fastq
ERR15466350	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466350_1.fastq
ERR15466351	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466351_1.fastq
ERR15466352	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466352_1.fastq
ERR15466353	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466353_1.fastq
ERR15466354	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466354_1.fastq
ERR15466355	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466355_1.fastq
ERR15466356	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466356_1.fastq
ERR15466357	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466357_1.fastq
```

e. Create combined txt file:
   i. cat manifest1.txt manifest2.txt > manifest.tsv
   ii. Move manifest.tsv to “microbiome” directory
   iii. The combined manifest.tsv file should look something like this:

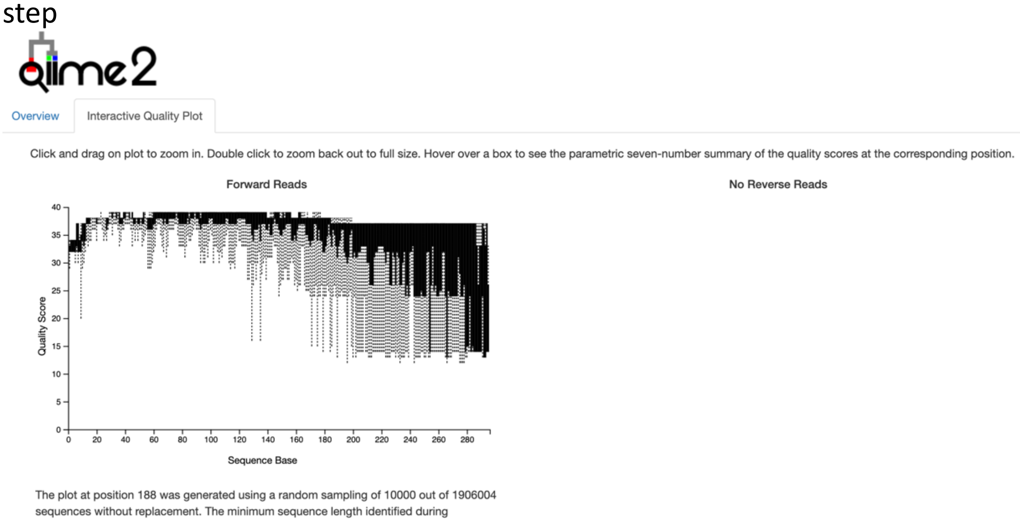
# single-end PHRED 33 fastq manifest file	
sample-id	absolute-filepath
ERR15466328	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466328_1.fastq
ERR15466329	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466329_1.fastq
ERR15466330	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466330_1.fastq
ERR15466331	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466331_1.fastq
ERR15466332	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466332_1.fastq
ERR15466333	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466333_1.fastq
ERR15466334	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466334_1.fastq
ERR15466335	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466335_1.fastq
ERR15466336	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466336_1.fastq
ERR15466337	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466337_1.fastq
ERR15466338	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466338_1.fastq
ERR15466339	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466339_1.fastq
ERR15466340	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466340_1.fastq
ERR15466341	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466341_1.fastq
ERR15466342	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466342_1.fastq
ERR15466343	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466343_1.fastq
ERR15466344	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466344_1.fastq
ERR15466345	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466345_1.fastq
ERR15466346	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466346_1.fastq
ERR15466347	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466347_1.fastq
ERR15466348	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466348_1.fastq
ERR15466349	/Users/stefankeller/desktop/single/data/microbiome/fastq/ERR15466349_1.fastq

   iv. Change back to parent (microbiome) directory

7. Begin Qiime2 import: At this point, analysis in Qiime2 is similar to that of the basic [Qiime2 tutorial;] just with the current dataset
    a. Activate conda environment:
        i. conda activate qiime2-amplicon-2024.10
            1. May need to change version number to fit qiime2 version downloaded
    b. Demultiplex dataset
        i. Import manifest that was created, and it will create a ".qza" file. Convert file to ."qzv" to visualize object



    c. View with function "qiime tools view name_of_file.qzv"
        i. This will open up a new internet tab for Qiime2 visualization for your data
    d. Go to "Interactive Quality Plot" to view data quality to set filter quality in next step



8. Sequence Filtering
    a. Can use Qiime2 deblur or dada2 function for filtering (this example is using DADA2)
        i. P-trim-left: trims on 5' end
        ii. P-trunc-len: trims 3' end
    b. In this example, a quality score of 30 or above was desired

```
(qiime2-amplicon-2024.10) Stefans-MBP-466:microbiome stefankeller$ qiime dada2 denoise-single \
>   --i-demultiplexed-seqs single-end-demux.qza \
>   --p-trim-left 0 \
>   --p-trunc-len 210 \
>   --o-representative-sequences rep-seqs.qza \
>   --o-table table.qza \
>   --o-denoising-stats stats-dada2.qza
Saved FeatureTable[Frequency] to: table.qza
Saved FeatureData[Sequence] to: rep-seqs.qza
Saved SampleData[DADA2Stats] to: stats-dada2.qza
(qiime2-amplicon-2024.10) Stefans-MBP-466:microbiome stefankeller$
```

9. Feature table and feature table summary

```
(qiime2-amplicon-2024.10) Stefans-MBP-466:microbiome stefankeller$ qiime feature-table summarize \
>   --i-table table.qza \
>   --o-visualization table.qzv \
>   --m-sample-metadata-file metadata.tsv
Saved Visualization to: table.qzv
(qiime2-amplicon-2024.10) Stefans-MBP-466:microbiome stefankeller$ qiime feature-table tabulate-seqs \
>   --i-data rep-seqs.qza \
>   --o-visualization rep-seqs.qzv
Saved Visualization to: rep-seqs.qzv
(qiime2-amplicon-2024.10) Stefans-MBP-466:microbiome stefankeller$
```

a. Table values: gives information about unique microbial sequences present in each sample, and frequency shows how often those microbial sequences appear in the sample

Table summary

Summary Statistic	Value
Number of samples	54
Number of features	1,844
Total frequency	1,762,936

Frequency per sample

	Frequency
Minimum frequency	3,111
1st quartile	25,688.5
Median frequency	32,493
3rd quartile	40,181
Maximum frequency	60,351
Mean frequency	32,647

b. Rep-seqs: representative clusters of similar microbial sequences

Sequence Table

To BLAST a sequence against the NCBI nt database, click the sequence and then click the *View report* button on the resulting page.

Download your sequences as a raw FASTA file

Click on a Column header to sort the table.

Feature ID	Sequence
0f6d36783ca5c49da26f66b4c0edf108	TACGTAGGGAGCGAGCGTTATCCGGATTTATTGGGTGTAAAGGGTGCGTAGACGGGAAGTCAAGTTAGTTGTGAAATCCCTCGGCTTAACTGAGGAACTGCAACTAAAACTGATTTT
c8599b3cdacef83db380153e3bfa491f	TACGTAGGGAGCAAGCGTTGTCCGGATTTACTGGGTGTAAAGGGTGCGTAGGCGGATCTGCAAGTCAGGCGTGAAATCCATGGGCTTAACCCATGAACTGCGCTTGAAACTGTGGGT
ce3122155169b5eaa19d2fb91a64afb7	TACGGAAGGTCCGGGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGCGGGAGATCAAGTCAGTTGTGAAAAGCAGCCGCTCAACGGTTGTCGTGCAGTTGATACTGGTTTT
45e12ad4f02a6b7232d5515be57e2ab7	TACGTAGGGGGCAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGTAGACGGCATAGCAAGTCTGAAGTGAAATACCCGGGCTCAACCTGGGAACTGCTTTGGAAACTGTTAAG
a4269bf2e8e74d1048a3eca4b370da76	TACGTAGGTCCCGAGCGTTGTCCGGATTTATTGGGCGTAAAGCGAGCGCAGGTGGTTTATTGAGTCTGGTGTAAAAGGCAGTGGCTCAACCATTGTATGCATTGGAAACTGGTAGAC
359ed3826827b24cf7f52143c86cc722	TACGTAGGGGGCTAGCGTTATCCGGATTTACTGGGCGTAAAGGGTGCGTAGGTGGTTTCTTAAGTCAGGAGTGAAAGGCTACGGCTCAACCGTAGTAAGCTCTTGAAACTGGGAAAC
6d40bd2ceef53e4ad112b1a758c009c1	TACGTAGGGGGCAAGCGTTGTCCGGAATGACTGGGCGTAAAGGGAGTGTAGGCGGCTTGGCAAGTTAGATGTGAAAGCCCGCAGCTCAACTGCGGAACTGCATCTAAAACTGTCAGG
4873c13639986cd21030fb416a71bf4c	TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTGCAGTTGAAACTGGCAGT
f0b1e2334808e74ded309c2eab755d63	TACGTAGGGAGCGAGCGTTGTCCGGAATTACTGGGTGTAAAGGGAGCGTAGGCGGGGTTGCAAGTTGAATGTGAAACCTACCGGCTTAACTGGTAGACTGCGTTCAAAACTGCAATT
66ba1553913520eb24813a78f909a455	TACGTAGGTGGCAAGCGTTGTCCGGAATTACTGGGTGTAAAGGGAGCGTAGGCGGGGATACAAGTTGAATGTTAAAACTATGGGCTCAACCCATAGTTGCGTTCAAAACTGTATTTC
6de89ed33478ebb11aa20ae2d8d1549f	TACGTAGGGAGCGAGCGTTGTCCGGAATTACTGGGTGTAAAGGGAGCGTAGGCGGGATCGCAAGTCAGATGTGAAAACTATGGGCTTAACCCATAAACTGCATTTGAAACTGTGGTT
e54e48be6ef6d850bba8270443e80d1b	TACGGAAGGTCCGGGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGCCGGAGATTAAGCGTGTTGTGAAATGTAGATGCTCAACATCTGCACTGCAGCGCGAACTGGTTTC

10. Taxonomic classification: This is a taxomonic database mapping the hypervariable regions to specific microbes. By aligning the sequences present in Rep-seqs to the database, taxonomic classification of the microbial population in each sample can be performed
    a. Import microbial reference/ classifier sequence
        i. wget \ -O "gg-13-8-99-515-806-nb-classifier.qza" \ "https://data.qiime2.org/classifiers/sklearn-1.4.2/greengenes/gg-13-8-99-515-806-nb-classifier.qza"
            1. Can also be customized to specific primer sequences used; this classifier seems to work with the 515F/806R primers used in this dataset
            2. Other classifier sequences (SILVA) also available
    b. Generate taxonomic profiles based on representative sequence data aligned to classifier

```
(qiime2-amplicon-2024.10) Stefans-MBP-466:microbiome stefankeller$ qiime feature-classifier classify-sklearn \
>   --i-classifier gg-13-8-99-515-806-nb-classifier.qza \
>   --i-reads rep-seqs.qza \
>   --o-classification taxonomy.qza
Saved FeatureData[Taxonomy] to: taxonomy.qza
(qiime2-amplicon-2024.10) Stefans-MBP-466:microbiome stefankeller$
```

c. Example of taxonomy table

Search:

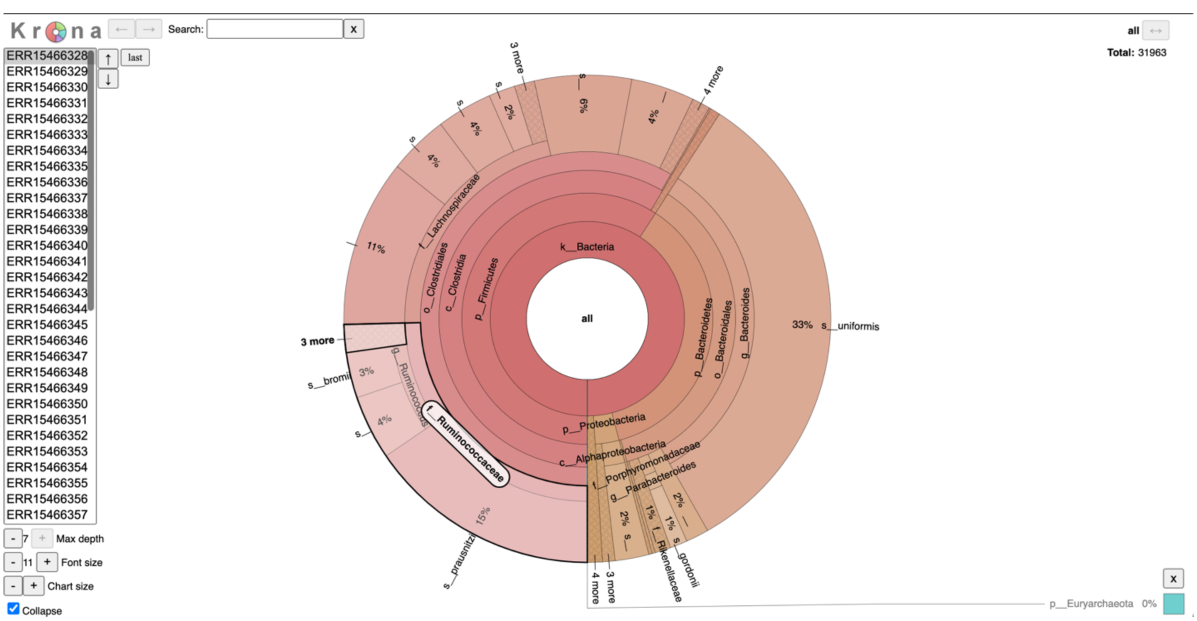
Feature ID #q2:types	Taxon categorical	Confidence categorical
00544a8c032d71ec4e38e36534904971	k__Bacteria; p__Proteobacteria; c__Gammaproteobacteria; o__Enterobacteriales; f__Enterobacteriaceae; g__Plesiomonas; s__shigelloides	0.9873206729474254
0063720ed811ba90158c3d7d199d5d3f	k__Bacteria; p__Firmicutes; c__Clostridia; o__Clostridiales; f__Lachnospiraceae	0.7769806359844554
0070f5028ce5f748387f82873a47a4f9	k__Bacteria; p__Firmicutes; c__Clostridia; o__Clostridiales; f__Ruminococcaceae; g__Ruminococcus; s__	0.9537853027231408
00a82e494013ca017f0e2035e61707df	k__Bacteria; p__Firmicutes; c__Clostridia; o__Clostridiales	0.7407908786134907
00d8e8792384c8a5d602061808250c3d	k__Bacteria; p__Firmicutes; c__Clostridia; o__Clostridiales	0.984071145874865
00fd7b4dbcc771a6101fc597acd71319	k__Bacteria; p__Firmicutes; c__Clostridia; o__Clostridiales; f__[Mogibacteriaceae]; g__; s__	0.9555036259045108
0144a2cc08e39804cabded49c3f1ea13	k__Bacteria; p__Firmicutes; c__Clostridia; o__Clostridiales; f__Ruminococcaceae; g__Oscillospira; s__	0.903250292381565
015fd4846f1548d1c1d14953bedbd9b1	k__Bacteria; p__Verrucomicrobia; c__Verrucomicrobiae; o__Verrucomicrobiales; f__Verrucomicrobiaceae; g__Akkermansia; s__muciniphila	0.9988823509855299
01a7010a0ff3e8a8cc52044db810d74d	k__Bacteria; p__Bacteroidetes; c__Bacteroidia; o__Bacteroidales	0.9969214223246359

11. Krona plot

a. Interactive plot to visualize microbial classification in individual sample

```
(qiime2-amplicon-2024.10) Stefans-MBP-466:microbiome stefankeller$ qiime krona collapse-and-plot \
> --i-table table.qza \
> --i-taxonomy taxonomy.qza \
> --o-krona-plot krona.qzv
Saved Visualization to: krona.qzv
```

Code for copying: qiime krona collapse-and-plot \
--i-table table.qza \
--i-taxonomy taxonomy.qza \
--o-krona-plot krona.qzv



If you would like to interact with this krona plot yourself, download the table.qza and taxonomy.qza file from here and run the code above! Alternatively, the krona plot can also be viewed by downloading the krona.qzv file directly and viewing with the qiime2 view command (code mentioned in section 7c)

Conclusion: For each sample, Qiime2 can create an interactive krona plot for microbial visualization and give an idea of relative abundance in each sample.

Citations

1. Bolyen, E., Rideout, J.R., Dillon, M.R. *et al.* Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. *Nat Biotechnol* **37**, 852–857 (2019). https://doi.org/10.1038/s41587-019-0209-9
2. Cassol, I., Ibañez, M. & Bustamante, J.P. Key features and guidelines for the application of microbial alpha diversity metrics. *Sci Rep* **15**, 622 (2025). https://doi.org/10.1038/s41598-024-77864-y